

Systems Engineering, Risk Management & Mission Assurance

How Quality Engineering helps control risk, manage changes, verify systems and improve mission reliability.

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Educational Note: This is an educational, illustrative case study based on publicly available systems-engineering and quality principles. It does not describe NASA's confidential or proprietary internal processes.

1. THE QUALITY CHALLENGE

A space mission is not made of one component. It is a combination of many connected systems such as:

Structure + Power + Communication + Sensors + Software + Thermal Systems + Control Systems

A problem in one subsystem can affect other parts of the mission. Therefore, Quality Engineering must look at the complete system, not only at individual components.

Component → Subsystem → Spacecraft → Mission

Systems Engineering: Means looking at the complete system and understanding how its different parts work together to achieve the mission objective.

2. RISK IDENTIFICATION

Before a mission moves forward, engineers need to ask: “What can go wrong?”

POSSIBLE RISKS INCLUDE

- Component failure or sensor error
- Software error / bus interruption
- Power interruption / Communication loss
- Manufacturing variation
- Incorrect design or specification
- Uncontrolled engineering changes

RISK ANALYSIS & APPROACH

Risk Analysis: Identifying possible problems early, understanding impact, and deciding priority.

Identify → Analyse → Control → Verify

3. FAULT TREE ANALYSIS — FTA

What is FTA? A method used to start with a major failure and work backwards to find its possible causes. Instead of asking “What component failed?” we ask “What different conditions could have caused this major failure?”

TOP EVENT: Spacecraft loses required function

- Power System Problem (OR) | • Communication System Problem (OR) | • Control System Problem

AND vs OR Logic: OR: Any one condition causes failure. | AND: Two or more conditions occur together.

Why FTA is useful: Helps engineers understand how major failures occur, identify failure paths, and focus on critical causes.

4. CONFIGURATION MANAGEMENT

Making sure that the correct and approved version of a design, drawing, software, specification or process is being used.

Design v1 → Change → Version 2 → Approval → Testing → Release

Why important? Prevents confusion between design, manufacturing, and testing teams through traceability.

5. VERIFICATION VS VALIDATION

These two words sound similar, but they answer different questions in aerospace quality engineering.

VERIFICATION

Question: “Did we build the product according to specified requirements?”

Checks requirements via inspection, analysis, calculation, testing.

→ Did we build it right?

VALIDATION

Question: “Does the product actually work for its intended purpose?”

Looks at system suitability for its intended operating environment.

→ Did we build the right thing?

6. TESTING & VALIDATION

Quality does not stop after manufacturing. Controlled cycle:

Requirement → Design → Build → Test → Analyse → Correct → Retest → Verify → Validate

Testing: Provides evidence of behavior under defined conditions. Problems identified must be: **Recorded** → **Investigated** → **Corrected** → **Retested**.

7. ANOMALY ANALYSIS

What is an anomaly? Something unexpected happens during testing or operation (behavior differs from prediction).

Quality Approach: Anomaly Detected → Record → Collect Evidence → Root Cause → Corrective Action → Retest → Effectiveness Check.

Important principle: Fixing the symptom is not enough; understand why it happened to prevent recurrence.

8. FINAL QUALITY ENGINEERING FRAMEWORK

NASA Mission Quality Approach:

System Requirements → Risk ID → Systems Eng → FTA → Config Control → Verification → Validation → Anomaly Analysis → Corrective Action → Retesting → Effectiveness → Mission Assurance

KEY TAKEAWAY

Quality Engineering in space is about understanding the complete system, identifying risks early, controlling changes, testing carefully, investigating anomalies, and ensuring mission readiness.

“Build it correctly, control every important change, test it with evidence, learn from failures, and continuously improve reliability.”